



Operating Instructions

Diesel engine-generator set
mtu 6R1600 DS300 / DG06R1600A1E
mtu 6R1600 DS330 / DG06R1600A2E

Controller MM7, MC7

SS180022/05E

System designation	Engine model	Application group
mtu 6R1600 DS300 / DG06R1600A1E	6R1600G10F	3B, continuous service, variable load, ICXN (PRP)
mtu 6R1600 DS300 / DG06R1600A1E	6R1600G70F	3D, emergency standby power, IFN (ESP)
mtu 6R1600 DS330 / DG06R1600A2E	6R1600G20F	3B, continuous service, variable load, ICXN (PRP)
mtu 6R1600 DS330 / DG06R1600A2E	6R1600G80F	3D, emergency standby power, IFN (ESP)

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Preface		6	Operation	
1.1	Preface	6		6.1	Preparations for putting into operation after extended out-of-service periods 45
2	Safety			6.2	Engine-generator set - Start 48
2.1	Important provisions for all products	7		6.3	Re-starting the engine following an automatic safety shutdown 49
2.2	Intended use of all products	8		6.4	Operational monitoring 50
2.3	Personnel and organizational requirements	9		6.5	Engine-generator set - Shutdown 51
2.4	Safety regulations for commissioning and operation	10		6.6	Emergency shutdown 52
2.5	Safety regulations for assembly, maintenance, and repair work	12		6.7	After shutdown - Putting engine-generator set out of operation 53
2.6	Fire and environmental protection, indirect materials	16		7	Maintenance Schedule
2.7	Standards for warning messages in the text and highlighted information	19		7.1	Preface 54
2.8	Overview of safety signs on engine-generator set	20		7.2	Maintenance task tolerances 55
2.9	Protective devices on engine-generator set - Overview	22		7.3	Maintenance schedule matrix 56
3	Transport			7.4	Maintenance tasks 57
3.1	Transport	24		8	Troubleshooting
4	Product Summary			8.1	Troubleshooting 58
4.1	Engine-generator set - General information	25		8.2	Fault messages 60
4.2	Engine-generator set - Overview	27		9	Task Description
4.3	Engine-generator set - Control versions	29		9.1	Engine-Generator Set 75
4.4	Engine	30		9.1.1	Engine-generator set - Test run 75
4.4.1	Engine side and cylinder designations	30		9.2	Engine 76
4.4.2	Engine - Overview	31		9.2.1	Engine - Barring with starting system 76
4.4.3	Sensors and actuators	33		9.3	Valve Drive 77
5	Technical Data			9.3.1	Valve clearance - Check and adjustment 77
5.1	Technical data mtu 6R1600 DS300, application group 3B (PRP)	35		9.3.2	Cylinder head cover - Removal and installation 79
5.2	Technical data mtu 6R1600 DS330, application group 3B (PRP)	38		9.4	Fuel System 80
5.3	Technical data mtu 6R1600 DS300, application group 3D (ESP)	40		9.4.1	Fuel system - Venting 80
5.4	Technical data mtu 6R1600 DS330, application group 3D (ESP)	42		9.5	Fuel Filter 81
5.5	Firing order	44		9.5.1	Fuel filter - Replacement 81
				9.5.2	Fuel prefilter - Draining 82
				9.5.3	Fuel prefilter - Replacement 83
				9.6	Air Filter 84
				9.6.1	Air filter - Replacement 84
				9.6.2	Air filter - Removal and installation 85
				9.7	Air Intake 86
				9.7.1	Service indicator - Signal ring position check 86
				9.8	Starting Equipment 87
				9.8.1	Starter - Condition check 87
				9.9	Lube Oil System, Lube Oil Circuit 88

9.9.1	Engine oil level – Check	88	10.2.16	Limit value table	185
9.9.2	Engine oil – Change	89	10.2.17	Messages table	186
9.10	Oil Filtration / Cooling	90	10.2.18	Programming tables (IO)	189
9.10.1	Engine oil filter – Replacement	90	10.2.19	Table from J1939	191
9.11	Coolant Circuit, General, High-Temperature Circuit	91	10.2.20	Selection table for the adaptation parameters	192
9.11.1	Engine coolant – Level check	91	10.2.21	Operating display of the control system	193
9.11.2	Engine coolant – Change	92	10.2.22	CAN communication	205
9.11.3	Engine coolant – Draining	93	10.2.23	Remote signaling	211
9.11.4	Engine coolant – Filling	94	10.2.24	CCRS – Communication option with modem	219
9.11.5	Engine coolant pump – Relief bore check	96	10.3	Automatic Transfer Switch (ATS) MC7	221
9.11.6	Coolant cooler – External contamination and leak-tightness check	97	10.3.1	Introduction of control system MC7 for the automatic transfer switch (ATS)	221
9.11.7	Coolant cooler – Cleaning	98	10.3.2	Dimensions, connection and installation dimensions	222
9.11.8	Preheater – Coolant line check	100	10.3.3	Front view of display unit	232
9.12	Belt Drive	101	10.3.4	Function types	235
9.12.1	Drive belt – Adjustment	101	10.3.5	Operation	237
9.12.2	Drive belt – Condition check	102	10.3.6	Control system – Inputs and outputs	238
9.12.3	Drive belt – Tension check	103	10.3.7	Control system – Messages	239
9.12.4	Drive belt – Replacement	104	10.3.8	Counter for maintenance and leasing	243
9.13	Wiring (General) for Engine/Gearbox/Unit	105	10.3.9	Time table	244
9.13.1	Engine cabling – Check	105	10.3.10	Measurement table	245
9.14	Accessories for (Electronic) Engine Governor / Control System	106	10.3.11	Setting table	246
9.14.1	Engine governor and connectors – Cleaning	106	10.3.12	Limit value table	247
9.14.2	Engine governor – Checking plug connections	107	10.3.13	Messages table	248
9.15	Generator	108	10.3.14	Programming table	250
9.15.1	Generator – Check	108	10.3.15	Synchronization table	251
9.15.2	Generator – Wiring check	109	10.3.16	Synchronization module table	252
9.15.3	Generator mounting – Securing screw firm seating check	110	10.3.17	Operating display of the control system	253
10	Monitoring and Control		10.4	CCLAN MODBUS Communication Module	263
10.1	General info, features and installation of the control units	111	10.4.1	Communication module CCLAN MODBUS – Introduction	263
10.2	MM7 Controller	112	10.4.2	Technical data	265
10.2.1	MM7 control system – Introduction	112	10.4.3	MODBUS protocol	267
10.2.2	Dimensions, connection and installation dimensions	114	10.4.4	Register mapping – Coil status	269
10.2.3	Front view of display unit	124	10.4.5	Register mapping – Input status	270
10.2.4	Function types	127	10.4.6	Register mapping – Input register	272
10.2.5	Operation	129	10.4.7	Register mapping – Holding register	275
10.2.6	Control system inputs and outputs	140	10.4.8	Status indicators	278
10.2.7	Control system – Messages	148	10.5	CCRS MODBUS Communication Module	279
10.2.8	Maintenance counters	163	10.5.1	Communication module CCRS MODBUS – Introduction	279
10.2.9	Options (extensions)	166	10.5.2	Technical data	280
10.2.10	Communication module	172	10.5.3	MODBUS protocol	282
10.2.11	Parameter table	179	10.5.4	Register mapping – Coil status	284
10.2.12	Time table	180	10.5.5	Register mapping – Input status	286
10.2.13	Measurement table	181	10.5.6	Register mapping – Input register	288
10.2.14	Setting table	182	10.5.7	Register mapping – Holding register	291
10.2.15	Table for analog transmitter and curve points	184	10.5.8	Status indicators	293
			11	Fluids and Lubricants Specifications	
			11.1	Preface	294
			11.1.1	General information	294
			11.2	Engine Oils	296

11.2.1	Requirements and oil change intervals	296	11.6.2	Multi-grade oils – Category 2.1 (Low SAPS oils), SAE grades 0W-30, 10W-30, 5W-40, 10W-40 and 15W-40	340
11.2.2	Viscosity grades	298	11.6.3	Multi-grade oils – Category 3, SAE grades 5W-30, 5W-40, 10W-40 and 15W-40 for diesel engines	344
11.2.3	Series-dependent usability of engine oils	299	11.6.4	Multi-grade oils – Category 3.1 (Low SAPS oils), SAE grades 5W-30, 10W-30 and 10W-40 for diesel engines	349
11.2.4	Used-oil analysis	300	11.6.5	Lubricating greases for diesel engine-generator set components	355
11.2.5	mtu Advanced Fluid Management System for engine oils – Test package for North America	302	11.7	Approved Coolants	356
11.3	Coolants	303	11.7.1	Antifreeze concentrates on ethylene glycol basis	356
11.3.1	Coolant – General	303	11.7.2	Antifreeze ready mixtures on ethylene glycol basis	359
11.3.2	Unsuitable materials in the coolant circuit	305	11.8	Flushing and Cleaning Specifications for Engine Coolant Circuits	361
11.3.3	Requirements imposed on freshwater	306	11.8.1	General information	361
11.3.4	Operational checks	307	11.8.2	Approved cleaning agents	362
11.3.5	Storage capability of coolant concentrates	308	11.8.3	Engine coolant circuits – Flushing	363
11.3.6	Color additives to detect leakage in the coolant circuit	309	11.8.4	Engine coolant circuits – Cleaning	364
11.3.7	Series-dependent usability of coolant additives	310	11.8.5	Cleaning engine coolant circuit assemblies	365
11.3.8	mtu Advanced Fluid Management System for coolant – Test package for North America	311	11.8.6	Coolant circuits contaminated with bacteria, fungi or yeast	366
11.4	Liquid Fuels	313	11.9	Cleaning	367
11.4.1	Diesel fuels – General information	313	11.9.1	General information	367
11.4.2	Biodiesel - biodiesel admixture	317	11.9.2	Approved cleaning agents	368
11.4.3	Heating oil EL	318	12	Appendix A	
11.4.4	Supplementary fuel additives	319	12.1	Abbreviations	369
11.4.5	Series-dependent approval of diesel fuel grades for mtu engines	321	12.2	Contact person/Service partner	372
11.4.6	Unsuitable materials in the diesel fuel circuit	326	13	Appendix B	
11.4.7	Measures prior to engine out-of-service periods >1 month	327	13.1	Standard Tools and Special Tools	373
11.4.8	mtu Advanced Fluid Management System for fuels – Test package for North America	328	13.2	Index	376
11.5	NOx Reducing Agent AUS 32 / AUS 40 for SCR Exhaust Gas Aftertreatment Systems	330			
11.5.1	General information and storage	330			
11.6	Approved Engine Oils and Lubricating Greases	331			
11.6.1	Multi-grade oils – Category 2 of SAE grades 10W-40, 15W-40 and 20W-40 for diesel engines	331			

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

The operating instructions are divided into the following chapters:

- Safety
- Transport
- Product overview
- Technical data
- Operation
- Maintenance Schedule
- Troubleshooting
- Task description
- Monitoring and control
- Fluids and Lubricants Specifications

Store these operating instructions where they can be accessed by all persons entrusted with operating or servicing the product.

Read these Operating Instructions carefully before starting any work. For safe operation of the product, always follow all safety instructions and work instructions.

Local accident prevention regulations and general safety regulations for the area of application of the product also apply.

These Operating Instructions form part of the product and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

2.4 Safety regulations for commissioning and operation

Safety regulations for commissioning

Install the product correctly and carry out acceptance in accordance with the manufacturer's specifications before commissioning the product. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Whenever the product is subsequently put into operation ensure that:

- All personnel is clear of the danger zone surrounding moving parts of the machine.
Electrically-actuated linkages may be set in motion when the Engine Control Unit (governor) is switched on.
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All protective devices are in place.
- All lines bearing media are ready for operation.
- The walkways in the operating room are unobstructed and safe.
- No persons wearing pacemakers or any other technical body aids are present.
- Adequate ventilation of the operating room is ensured for any operating status.
- In the first few hours of operation, the product emits gases as a result of smoldering e.g. lacquers or oil.
These gases may be hazardous to health. Always wear respiratory protection in the operating room during this period.
- The exhaust system is leak-tight and the gases are vented to atmosphere.
- The product is free of any damage, this applies in particular to lines and cabling.
- All components are properly grounded. Ground separately by means of a grounding stake as necessary.
- The power supply (mains plug) is always freely accessible to facilitate disconnecting the device or system from the power supply at any time.
- Battery terminals, generator terminals or cables are protected against accidental contact.
- All connections have been correctly allocated e.g. +/- polarity, fuel line/reducing agent line, supply/return.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Smoking is prohibited in the area of the product.

Safety regulations during operation

The operator must be familiar with the control and display elements.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

Practice emergency procedures, especially emergency stopping, at regular intervals.

Take the following steps if any system malfunctions are detected or signaled by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- Respond by taking any necessary emergency action, e.g. emergency stop.

After a safety shutdown, the engine may only be started after the cause of the shutdown has been eliminated.

Contact Service if the root cause of the malfunction cannot be clearly identified.

Operation

Do not remain in the operating room when the product is running unless absolutely necessary. Keep your stay as short as possible.